Childhood soy intake and breast cancer risk in Asian American women.

INTRODUCTION: Historically, breast cancer incidence has been substantially higher in the United States than in Asia. When Asian women migrate to the United States, their breast cancer risk increases over several generations and approaches that for U.S. Whites. Thus, modifiable factors, such as diet, may be responsible. METHODS: In this population-based case-control study of breast cancer among women of Chinese, Japanese, and Filipino descent, ages 20 to 55 years, and living in San Francisco-Oakland (California), Los Angeles (California) and Oahu (Hawaii), we interviewed 597 cases (70% of those eligible) and 966 controls (75%) about adolescent and adult diet and cultural practices. For subjects with mothers living in the United States (39% of participants), we interviewed mothers of 99 cases (43% of eligible) and 156 controls (40%) about the daughter's childhood exposures. Seventy-three percent of study participants were premenopausal at diagnosis. RESULTS: Comparing highest with lowest tertiles, the multivariate relative risks (95% confidence interval) for childhood, adolescent, and adult soy intake were 0.40 (0.18-0.83; P(trend) = 0.03), 0.80 (0.59-1.08; P(trend) = 0.12), and 0.76 (0.56-1.02; P(trend) = 0.04), respectively. Inverse associations with childhood intake were noted in all three races, all three study sites, and women born in Asia and the United States. Adjustment for measures of westernization attenuated the associations with adolescent and adult soy intake but did not affect the inverse relationship with childhood soy intake. DISCUSSION: Soy intake during childhood, adolescence, and adult life was associated with decreased breast cancer risk, with the strongest, most consistent effect for childhood intake. Soy may be a hormonally related, early-life exposure that influences breast cancer incidence.